

DermaNet: Flexible-View Fusion for Dermoscopic–Clinical Skin Lesion Classification

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Skin Lesion Classification

ISIC2019 and MILK10 datasets:

- **Fine-grained classification:** Small images with limited spatial structure
- **Imbalanced classes:** Rare classes amount to 1-5% of the dataset
- **Out-of-Distribution (OOD) detection:** Both datasets have samples outside the known classes
- **Uncertainty estimation:** Important for medical applications and OOD detection
- **Two views/modalities:** Paired dermoscopic and clinical images in MILK10
- **Explainability:** Identifying clinical features, displaying attention maps, etc.



Dermoscopic Image



Clinical Image

Comparison of Leading Deep Learning Architectures on Skin Lesion Classification Using the ISIC 2019 Dataset

Baseline Architectures and Ensemble Models

- Comparison of **EfficientNet-B5**, **ViT**, **MambaVision**
- Focal loss, test-time augmentation, and hyperparameter optimization
- Heterogeneous ensemble of complementary architectures

→ **74.3% Macro-F1** → **74.8% Accuracy**

Source: Aktan et al., "Comparison of Leading Deep Learning Architectures on Skin Lesion Classification Using the ISIC 2019 Dataset"
TIPTEKNO 2025, Gazi Mağusa, Türkiye.

Exponential Moving Average of Vision Transformers

Reducing parameter fluctuations during convergence

- Vision Transformer training on **ISIC2019**
- Parameter instability from small-batch training
- Exponential Moving Average (**EMA**)

→ **+6.93% Macro-F1** → **+2.67% Accuracy**

EMA reduces parameter variability and improves performance with **essentially no additional inference cost**.

Source: Şen et al., "Exponential Moving Average of Vision Transformers for Skin Lesion Classification," SIU 2026, İstanbul, Türkiye.

Improving Skin Lesion Classification Using Metadata

Incorporating the meta-data

- Age, gender, anatomical site, etc.
- **MONET descriptors** of the image
- Modality-specific metadata

→ **56.31% Macro-F1** → **+3.7% vs. baseline**

Source: Oğuz et al., "Improving Skin Lesion Classification Using Metadata," TIPTEKNO 2026, Gaziantep, Türkiye.

Uncertainty-Aware Skin Lesion Classification with Evidential Error-Correcting Output Codes

Uncertainty estimation via Evidential Deep Learning

- Evidential Deep Learning (**EDL**)
- Error-Correcting Output Codes (**ECOC**)
- Uncertainty estimation across binary dichotomies
- Out-of-dichotomy regularization for **OOD detection**

→ **63.8% Macro-F1** → **69.7% OOD AUROC**

Source: Tuygan et al., "Uncertainty-Aware Skin Lesion Classification with Evidential Error-Correcting Output Codes," TIPTEKNO 2026, Gaziantep, Türkiye.

Dataset 1: ISIC2019

Dataset

- **Dermoscopic** images only
- 8 diagnostic classes
- 25,331 training images
- Official held-out test set with an *UNKNOWN* class

Class	Training %
NV (Melanocytic nevus)	50.8%
MEL (Melanoma)	17.9%
BCC (Basal cell carcinoma)	13.1%
BKL (Benign keratosis)	10.4%
AK (Actinic keratosis)	3.4%
SCC (Squamous cell carcinoma)	2.5%
VASC (Vascular lesion)	1.0%
DF (Dermatofibroma)	0.9%

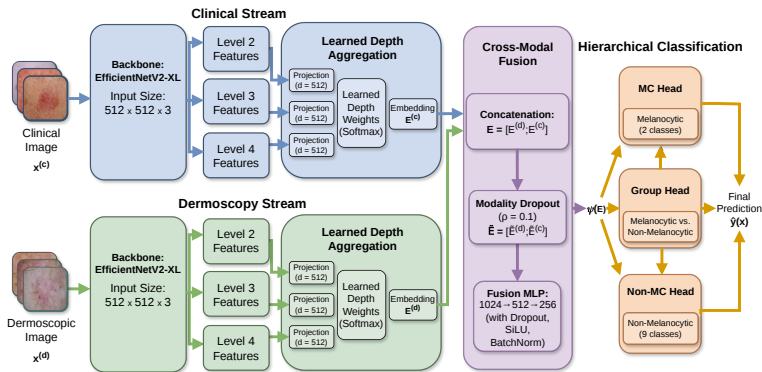
Dataset 2: MILK10k

Dataset

- Paired **clinical** + **dermoscopic** images
- 11 diagnostic classes
- 5,240 training lesions
(10,480 images)
- Official held-out test set including two *OTHER* classes

Class	Training %
BCC (Basal cell carcinoma)	48.1%
NV (Melanocytic nevus)	14.2%
BKL (Benign keratinocytic lesion)	10.4%
SCC/KA (Squamous cell carcinoma / KA)	9.0%
MEL (Melanoma)	8.6%
AK/IEC (Actinic keratosis / IEC)	5.8%
DF (Dermatofibroma)	1.0%
INF (Inflammatory / infectious)	1.0%
VASC (Vascular lesion)	0.9%
BEN_OTH (Other benign)	0.8%
MAL_OTH (Other malignant)	0.2%

DermaNet: Overall Architecture



Multi-Level
Features

Paired Feature
Fusion

Hierarchical
Prediction

Flexible View
(Paired or Single)

1. Multi-Level Representation Learning

Multilevel Features

We extract features from three depths:

$$L_2, \quad L_3, \quad L_4$$

Each is projected to a common 512-dimensional representation e_i , $i \in 2, 3, 4$.

Learned depth aggregation

The contribution of each depth is learned:

$$\alpha = \text{softmax}(w)$$

$$E = \alpha_2 e_2 + \alpha_3 e_3 + \alpha_4 e_4$$

Instead of manually deciding which feature level matters, the model learns the weighting during training.

→ + 0.90 Macro-F1 in MILK10 → +1.55 Micro-F1 in ISIC2019

What Feature Depths Did DermaNet Learn to Use?

Clinical stream

Feature Level	Learned Weight
Level 2	0.078
Level 3	0.203
Level 4	0.719

Dermoscopic stream

Feature Level	Learned Weight
Level 2	0.086
Level 3	0.222
Level 4	0.692

Both streams rely most strongly on the **deep semantic representation**, while still retaining information from intermediate depths.

Source: Learned depth weights of the MILK10k model.

2. Clinical–Dermoscopic Feature Fusion

Separate view representations

After multi-level aggregation:

$$E^{(c)} \in \mathbb{R}^{512}$$

for the clinical image, and

$$E^{(d)} \in \mathbb{R}^{512}$$

for the dermoscopic image.

Joint representation

The two embeddings are concatenated:

$$E = [E^{(c)}; E^{(d)}] \in \mathbb{R}^{1024}$$

and processed by the fusion network:

$$1024 \rightarrow 512 \rightarrow 256$$

The final 256-dimensional representation is shared by all classification heads.

3. Flexible-View Learning with Modality Dropout

MILK10k provides paired views, but other datasets such as ISIC2019 provide only dermoscopic images. During training, DermaNet randomly masks views from paired samples:

Training Condition	Clinical	Dermoscopy	Probability
Both views	✓	✓	80%
Clinical only	✓	×	10%
Dermoscopy only	×	✓	10%

→ **Single network operates with paired, clinical only, or dermoscopy only views.**

4. Clinically Structured Hierarchical Classification

Hard decision

DermaNet first predicts:

Melanocytic vs. Non-Melanocytic

Soft hierarchy

Two separate classifiers predict the diagnosis in each subgroup.

$$P(y|x) = P(g(y)|x)P(y|x, g(y))$$

Inference Method	ISIC2019 Micro-F1
Soft hierarchy	77.40%
Hard hierarchy	77.42%

Melanocytic lesions originate from melanocytes, the pigment-producing cells of the skin. In our hierarchy, this group contains **MEL** and **NV**; the remaining classes are grouped as non-melanocytic.

Contribution of Each Component

Configuration	MILK10k		ISIC2019	
	Macro-F1	Δ	Micro-F1	Δ
Baseline EfficientNetV2	40.3%	–	73.63%	–
+ Multi-Level Aggregation	41.2%	+0.9	75.18%	+1.55
+ Hierarchical Prediction	45.4%	+4.2	77.42%	+2.24
+ Paired Feature Fusion	49.7%	+4.3	–	–

MILK10k

Largest gains came from **paired feature fusion** and **hierarchical prediction** and **hierarchical prediction**.

40.3% → **49.7%**

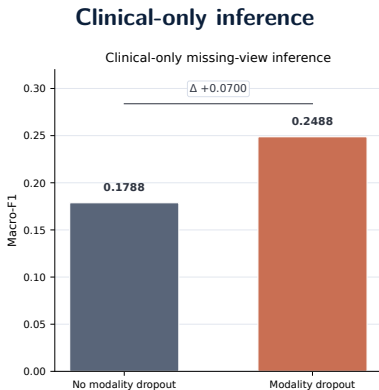
ISIC2019

Multi-level aggregation and **hierarchical prediction** both improved performance..

73.63% → **77.42%**

Missing-View Robustness with Modality Dropout

Robustness ablation: trained on mixed MILK10k + ISIC data and evaluated on held-out paired samples



When dermoscopic image is unavailable

Clinical image only

Without modality dropout

0.179



With modality dropout

0.249

+0.070 Macro-F1

Computational Overhead

Parameters

Model	Parameters
Score-level baseline	412.05M
DermaNet	413.72M

+0.4%

Estimated compute

Model	GMACs / Lesion
Score-level baseline	187.272
DermaNet	187.275

Essentially unchanged

ISIC2019: Comparison with Prior Work

70:10:20 Train–Validation–Test Split

Method	Micro-F1	Macro-F1	AUC	Bal. Acc.
Sun et al. (2021)	–	–	91.5%	66.2%
Yao et al. (2022)	75.0%	–	–	–
Li et al. (2022)	75.1%	–	–	–
Alsahafi et al. (2023)	71.3%	–	–	–
Liu et al. (2026)	79.9%	68.9%	–	–
DermaNet	82.6%	72.7%	95.4%	70.8%

DermaNet achieves strong performance across all reported metrics under the same 70:10:20 evaluation setting.

Source: DermaNet Table 2; literature comparison using the 70:10:20 train–validation–test split.

DermaNet: Main Takeaways

1. **Multi-level representation learning:** DermaNet learns how much information to use from different EfficientNetV2 depths.
2. **Feature-level multimodal fusion:** Clinical and dermoscopic information are combined before the final classification decision.
3. **Flexible-view learning:** Modality dropout enables the same model to work with paired and single-view inputs.
4. **Clinically structured prediction:** A melanocytic / non-melanocytic hierarchy improves fine-grained classification.

MILK10k

49.7%

Macro-F1

ISIC2019

77.42%

Micro-F1

Overhead

+0.4%

parameters

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